REPORT 01 / ANALYSIS / 12 NOVEMBER 2026

SCROLL HERO FRAME.

How the “Ideas are human” founder section on grubogrubo.com works, measured element by element, and the plan to rebuild it as a reusable frame you can drop into any web project.

REFERENCE	grubogrubo.com / section #team.founder-cinematic
SOURCE FILES READ	founder-scrub.js (v36), reveal-motion.js (v20), live CSS rules
MEASURED AT	1600 x 832 viewport (your screenshots: same view at 125% scaling)
STATUS	Analysis complete. Build NOT started: waiting for your go-ahead.

Contents

#	Section	What it covers
1	At a glance	what the frame is and the key numbers
2	Storyboard	your 8 screenshots placed on the scroll timeline
3	How the scroll works	pinning, the scrub formula and the two phases
4	Anatomy	layers, HTML structure, measured layout
5	Design tokens	colours, type scale, spacing, gradients
6	Execution flow	step-by-step logic of both scripts, motion curves
7	Responsive behaviour	what changes at 820 / 550 / 390px
8	The video asset	requirements and why it must be re-encoded
9	Reusable package plan	files, class names, variables, how to drop it in
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A	Glossary	plain-language meaning of the technical terms

Companion files for AI tools (Markdown, in the `prompts/` folder next to this report):
`01-build-scroll-hero-frame.md` is a complete, self-contained build prompt with the full spec and acceptance tests;
`02-generate-scroll-video.md` is a prompt for producing your own background footage and encoding it for scrubbing.

1. At a glance

The section is a **scroll-driven cinematic frame**. A full-screen video sits pinned behind the content and never plays by itself: the page's scroll position *is* the video's playhead. Scroll down and the film advances (the founder, photo cards orbiting him, light streaks); scroll up and it rewinds. On top, a giant two-line headline, a lead paragraph and a frosted-glass profile card scroll past, each block fading in with a soft blur-and-rise as it reaches the screen. A 3px purple bar along the bottom edge shows how far through the film you are.

Everything is plain HTML, CSS and about 80 lines of JavaScript. No libraries (no GSAP, no ScrollTrigger, no Lottie). That is what makes it easy to lift into other projects.

Key figures (measured on the live site)

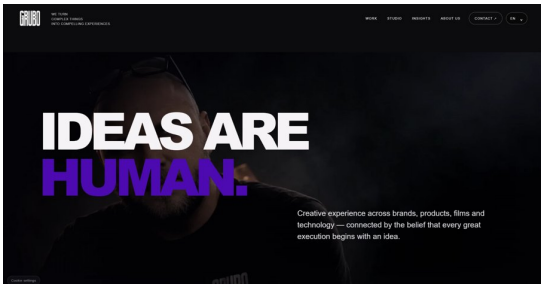
Item	Value	Note
Section height	min 175vh; measured 2106px = 253vh	content (the tall card) makes it taller than the minimum
Video	7.04 s, 1280 x 720, muted, inline	never plays; only currentTime is set
Pin point	at 40% scroll progress	section top reaches the top of the screen
Progress bar	3px, #4C0AAF on 15% white	width = scroll progress
Reveal motion	rise 27px + blur 5px + fade, 0.9-1.0 s	stagger 0/75/150/225/300 ms
Breakpoints	820px, 550px, 390px	below 820px the layout is rebuilt as a column
Fonts	Arial Black + Arial (system)	no web-font download
Dependencies	none	vanilla JS, IntersectionObserver, requestAnimationFrame

What you will get when the build is approved

- **scroll-hero.css**: every rule from the source, renamed to a safe `.shf-` prefix and driven by CSS variables.
- **scroll-hero.js**: the scrub engine and the reveal engine merged, working for any number of frames per page.
- **index.html**: a demo page matching your screenshots (with a simple header), with placeholder text and a slot for your video.
- **README.md**: how to drop the frame into another project, rename text, swap colours and video.
- A **Report 02 (build)** PDF with screenshots of the finished frame next to the reference, plus updated AI prompt files.

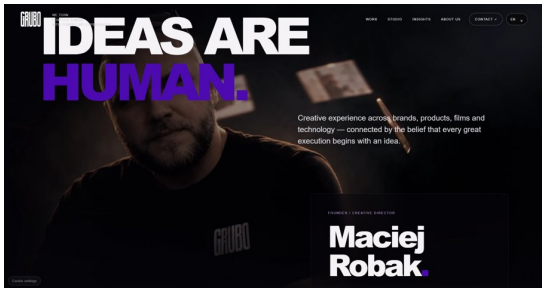
2. Storyboard: your 8 screenshots

Each screenshot was placed on the timeline by measuring where known elements sit on screen and converting that to scroll progress (screenshots are 2000px wide = the 1600px layout at 125% Windows scaling). Video time = progress x 7.04 s.



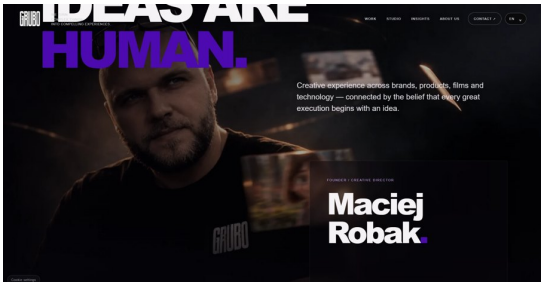
1 **33% / 2.3 s**

Section top has entered (dark band above is the previous section). Headline and lead are visible; the video is still in its dark opening.



2 **47% / 3.3 s**

Just after the pin. Headline rising; the glass card arrives bottom-right with kicker and name. Founder lit, first cards in the air.



3 **52% / 3.7 s**

Headline leaving the top. Cards begin to orbit; the lead overlaps the header area.



4 **56% / 4.0 s**

The 20+ counter has revealed inside the card; more cards and streaks.



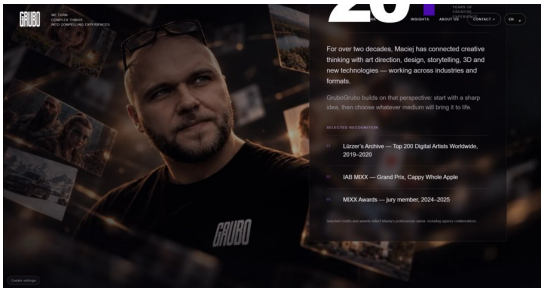
5 **61% / 4.3 s**

Lead under the navigation; counter rule line visible. The camera starts pushing in.



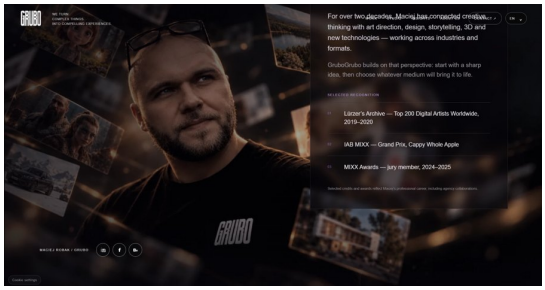
6 **75% / 5.2 s**

Card fills the right column: name, 20+, bio and second bio. Video near full brightness.



7 **94% / 6.6 s**

Awards list 01-03 and the credit note. Full ring of cards around the subject.



8 **99% / 7.0 s**

End of the film. Signature row with round social buttons bottom-left; card bottom edge visible.

3. How the scroll works

The whole effect rests on one CSS trick and one formula.

The CSS trick: a sticky backdrop that takes up no space

The backdrop (video + shade + progress bar) is `position: sticky; top: 0; height: 100vh` and has `margin-bottom: -100vh`. The negative margin cancels its own height, so the content that follows starts at the very top of the section and scrolls *over* the video, while the video stays glued to the screen until the section runs out.

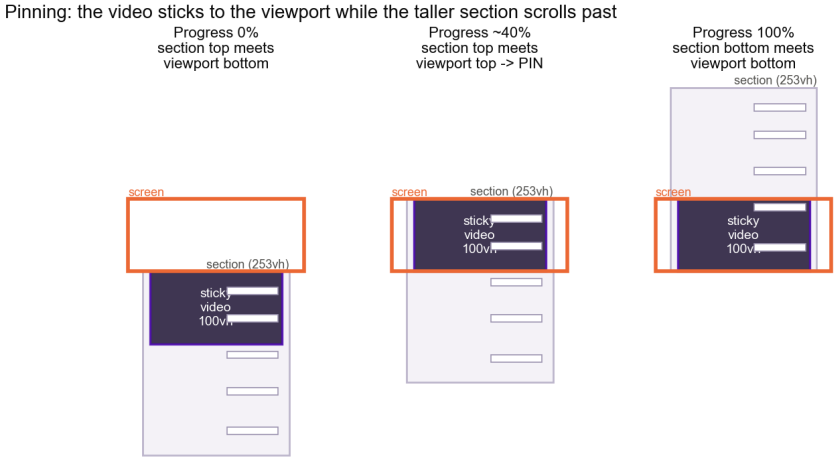


Figure 3.1 The three key moments. Orange = your screen. Dark block = the sticky video. White bars = content blocks scrolling past.

The formula: scroll progress becomes video time

```
progress = clamp( (viewportHeight - section.top) / section.height , 0 , 1 )
video.currentTime = progress x video.duration
progressBar.width = progress x 100%
```

Progress is 0 when the section's top edge appears at the bottom of the screen, and 1 when the section's bottom edge reaches the bottom of the screen (the moment the next section starts to show). That gives two phases: in **Phase A** the section is still sliding in and the video moves with it; in **Phase B** the video is pinned and the text slides over it. Because the formula divides by the section's real height, the film always finishes exactly when the section ends, whatever the screen size or text length.

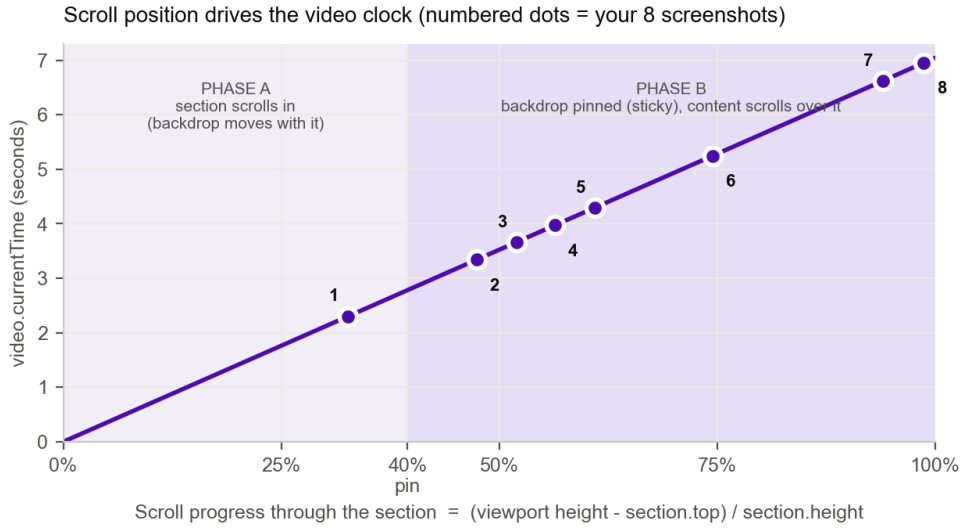


Figure 3.2 Measured mapping at 1600 x 832. The relationship is a straight line: equal scroll distance = equal film time.

4. Anatomy

Exploded view of the frame (front = top)

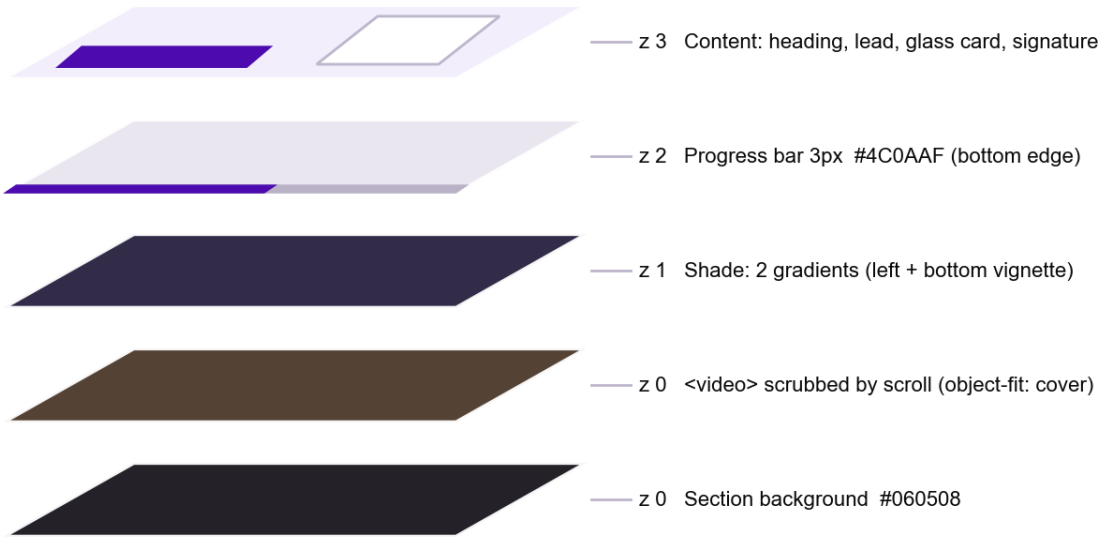


Figure 4.1 Stacking order. The shade sits between video and text so white type stays readable over any frame.

HTML structure (as it will be in the reusable version)

```
<section class="shf" data-scroll-hero>
  <div class="shf-backdrop" aria-hidden="true">        <- sticky, 100vh, margin-bottom:-100vh
    <video class="shf-video" src="your.mp4" poster="your.jpg"
           muted playsinline preload="auto"></video>
    <div class="shf-shade"></div>                      <- 2 gradients
    <div class="shf-progress"><span></span></div>      <- 3px bar
  </div>
  <div class="shf-content">                            <- z 3, flex column
    <div class="shf-heading">
      <h2>IDEAS ARE<br><em>HUMAN.</em></h2>
      <p class="shf-lead">...</p>
    </div>
    <div class="shf-grid">                              <- pushes card right
      <div class="shf-card">                            <- glass card
        <div class="shf-kicker">FOUNDER / CREATIVE DIRECTOR</div>
        <h3>Name<span class="shf-dot">.</span></h3>
        <div class="shf-counter"><strong>20<span>+</span></strong><span>YEARS OF ...</span></div>
        <p class="shf-bio">...</p>  <p class="shf-bio shf-bio-2">...</p>
        <div class="shf-awards"> label + .shf-award rows + .shf-note </div>
      </div>
    </div>
    <div class="shf-signature">NAME / BRAND  + round .shf-social links</div>
  </div>
</section>
```

First screen of the frame (desktop, > 820px): measured layout



Figure 4.2 First screen on desktop. Content column is max 1600px, centred, with 6.5vw side padding and 15vh top padding.

5. Design tokens

Colours

	Hex	Role	Used for
	`#4C0AAF`	Accent	HUMAN., dots, the +, progress bar, button hover
	`#060508`	Section background	behind everything
	`#F5F3F6`	Heading text	h2, h3
	`#E0DCE4`	Lead text	with shadow 0 2px 25px #000a
	`#D2CCD5`	Bio	first bio paragraph
	`#979198`	Bio (second)	dimmer follow-up paragraph
	`#E7E4E8`	Award text	award rows
	`#8E898F`	Credit note	11px fine print
	`#C3A0EC`	Kicker / labels	10px uppercase labels
	`#9165CA`	Award numbers	01 / 02 / 03
	`#3C393F`	Rules	counter underline; award rows use #3D3840

Glass card: background `rgba(7,6,12,.52)`, border 1px `rgba(255,255,255,.11)`, `backdrop-filter: blur(10px)`, radius 2px. Social buttons: 42px circles, 1px border at 65% white, background `rgba(9,9,11,.76)` + blur 9px; on hover they turn accent purple and lift 3px.

Shade over the video

```
linear-gradient(90deg,  rgba(5,4,9,.77) 0%, rgba(5,4,9,.46) 43%, rgba(5,4,9,.21) 76%, rgba(5,4,9,.46) 100%),
linear-gradient(0deg,   rgba(5,4,9,.78), transparent 28%, transparent 68%, rgba(5,4,9,.44))
video:  object-fit: cover;  object-position: center 42%;  filter: contrast(1.05) saturate(.88)
```

The first gradient darkens the left (behind the headline) and right edge; the second darkens the bottom (behind the signature) and the top (behind the site header). The middle-right stays brightest, which is where the subject's face is lit.

Typography

Element	Font	Size	Letter-spacing / line-height
Headline h2	Arial Black	clamp(66px, 9.4vw, 175px)	-0.086em / 0.9
Lead	Arial	clamp(18px, 1.5vw, 24px)	normal / 1.5, max-width 660px
Name h3	Arial Black	clamp(44px, 5.7vw, 94px)	-0.07em / 0.96
Counter 20+	Arial Black	clamp(110px, 14vw, 220px)	-0.1em / 0.8
Kicker, labels	Arial	10px uppercase	0.18em
Bio / second bio	Arial	clamp(16px,1.35vw,21px) / clamp(15px,1.2vw,19px)	1.6
Award rows	Arial	clamp(14px, 1.2vw, 18px)	1.45, grid 40px + 1fr
Credit note	Arial	11px	1.65, max-width 510px
Signature	Arial	10px	0.17em

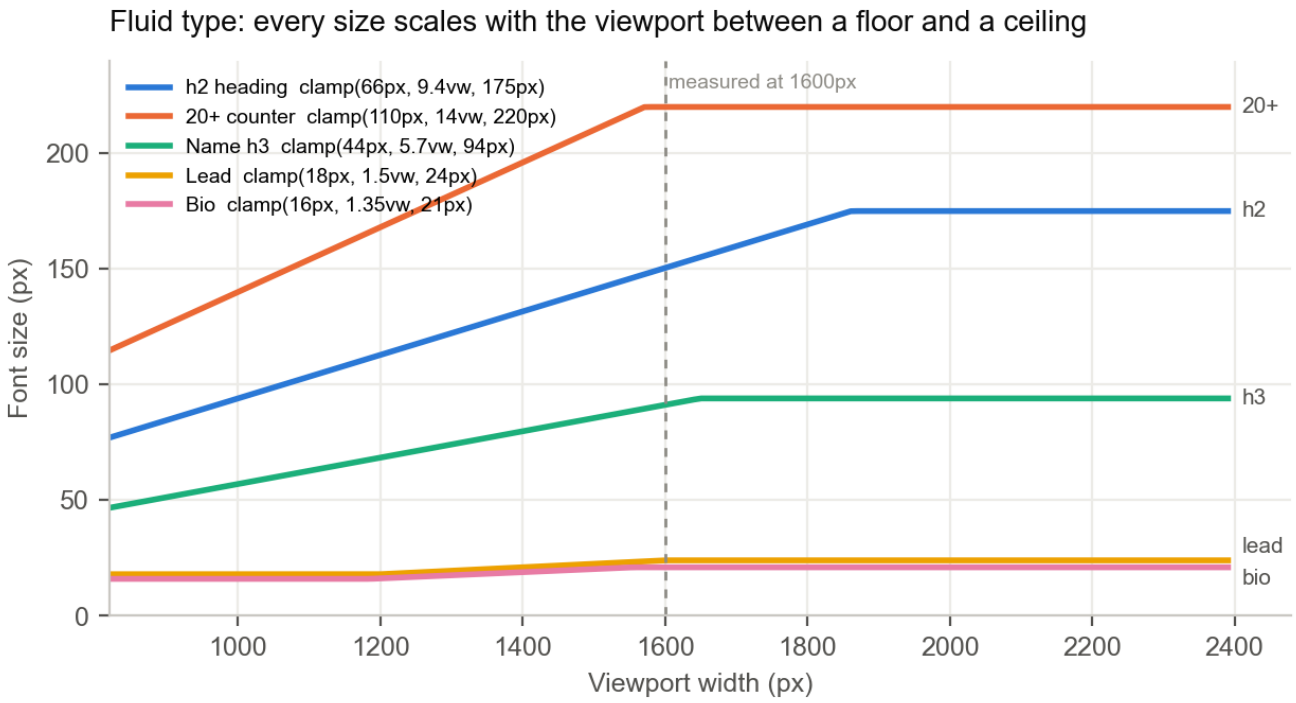


Figure 5.1 clamp(min, preferred, max): sizes grow with the window, but never below the floor or above the ceiling.

Spacing

Where	Value
Content padding	15vh top, 6.5vw sides, 14vh bottom; max-width 1600px centred
Heading block	h2 margin 35px 0 28px; lead margin-bottom 75px; heading group margin-bottom 8vh
Glass card	width min(100%, 620px); padding clamp(24px, 3.2vw, 54px)
Inside card	name margin 20px 0 22px; counter gap 23px, padding-bottom 32px + rule; bio top 34px; second bio 20px; awards 42px
Award rows	22px vertical padding, 1px top rule
Signature row	12vh above, 24px gap, social buttons 9px apart

6. Execution flow

Two independent scripts run on the page. Neither needs the other, and both only read the page; they never fetch anything.

6.1 The scrub engine (founder-scrub.js)

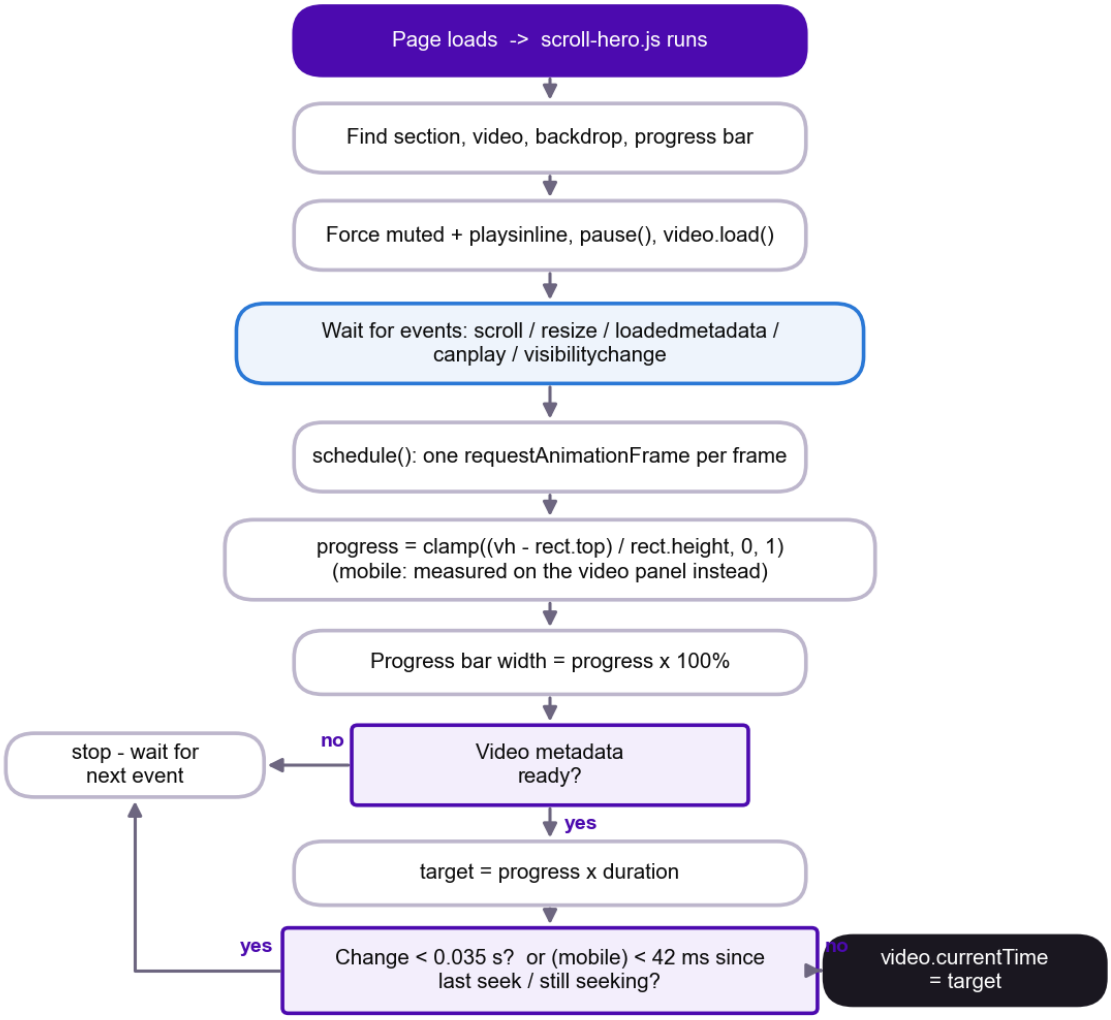


Figure 6.1 Every scroll event is folded into at most one calculation per screen refresh.

Why the guards matter

- **requestAnimationFrame batching**: a fast trackpad fires dozens of scroll events per frame. Only the last one per frame is used.
- **0.035 s minimum change**: tiny scroll nudges do not trigger a seek, so the decoder is not asked for the same frame twice.
- **Mobile throttle (42 ms / skip while seeking)**: phones decode slower; piling up seeks makes the video freeze, so it waits.
- **iOS priming**: Safari will not draw a frame of a video that was never started. On the first touch the script calls `play()` and immediately `pause()`, muted, which unlocks frame drawing.
- **'play' listener**: if anything tries to play the video (autoplay, extensions), it is paused again at once.

6.2 The reveal engine (reveal-motion.js)

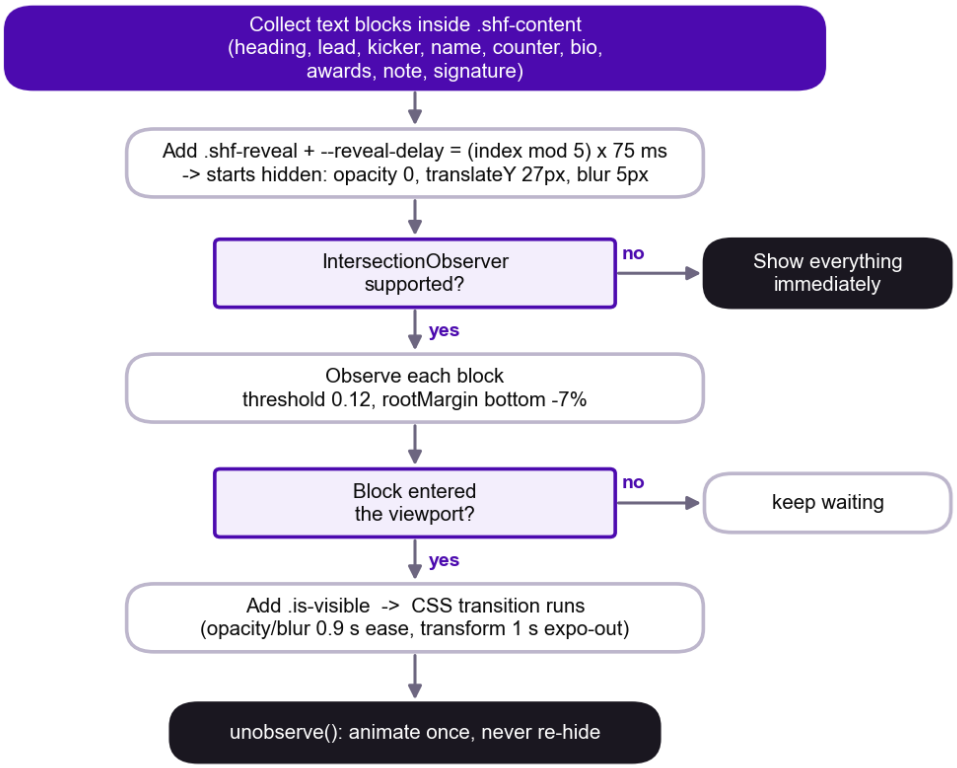


Figure 6.2 Each block animates exactly once, the first time it enters the screen.

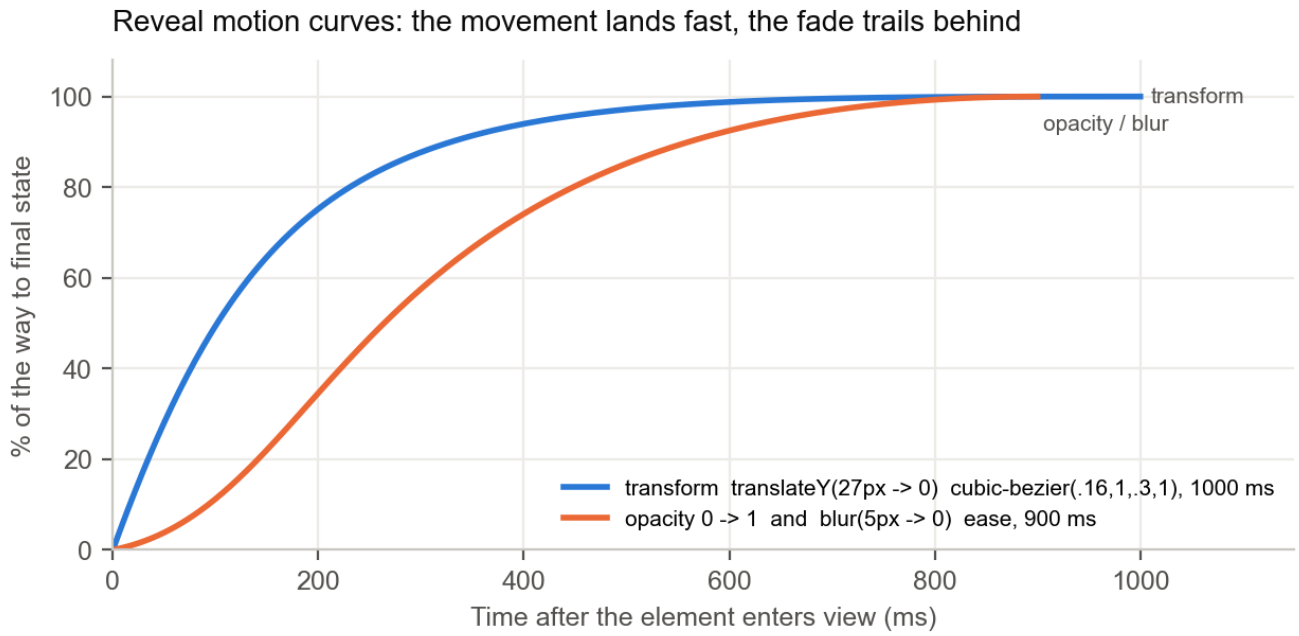


Figure 6.3 The rise uses a strong ease-out, so it lands almost immediately; the fade and blur trail behind it, which gives the soft 'focus-pull' feel.

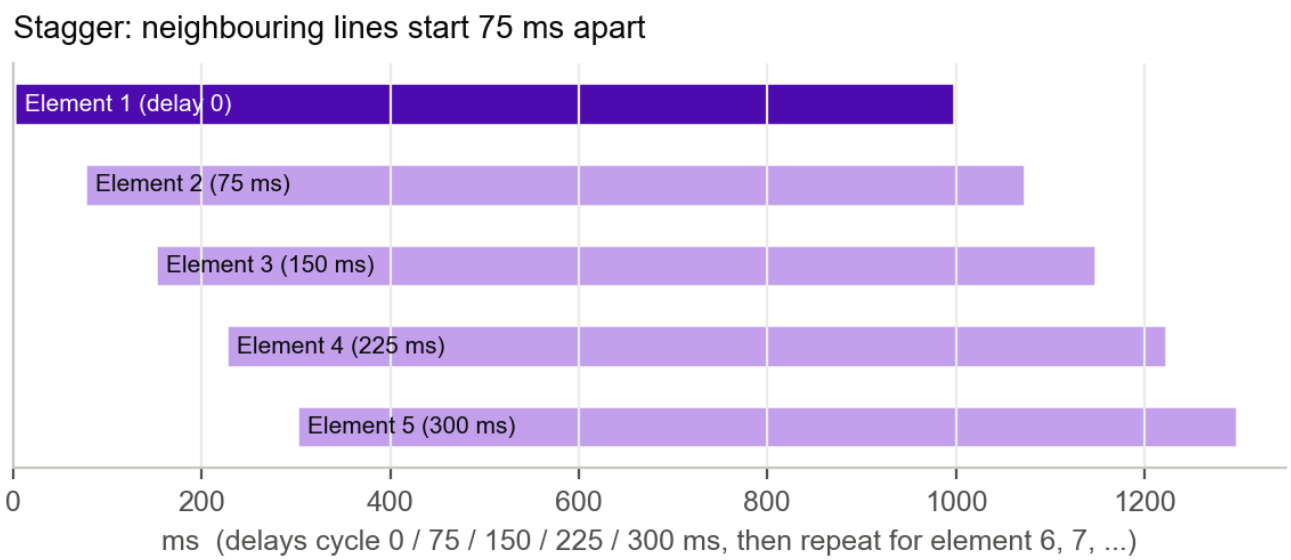


Figure 6.4 Delays repeat every five elements, so a long list never waits more than 300 ms.

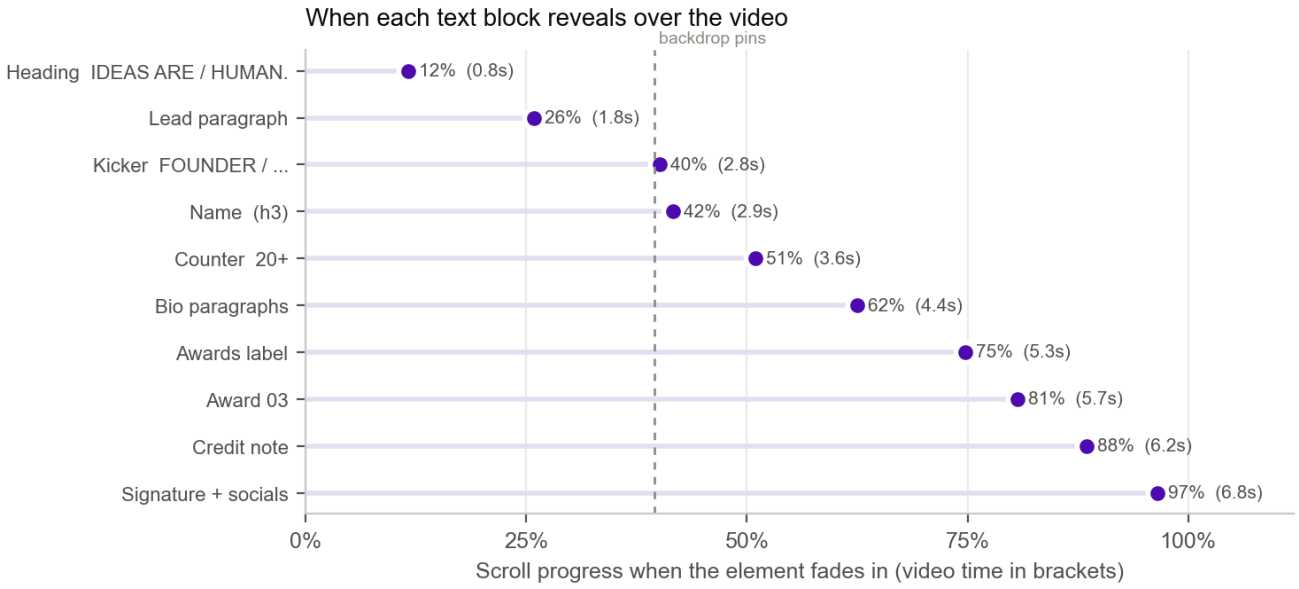


Figure 6.5 Measured reveal points. Reveals are tied to scroll position, not the clock, so they line up with the film on every visit.

7. Responsive behaviour



Above 820px the frame is the cinematic version described so far. At 820px and below the source does something clever: it sets `display: contents` on the content wrappers, which makes the wrappers disappear from layout while keeping their children. The children (heading, video, lead, card contents, signature) can then be re-ordered with `order` into one simple column. The video stops being a full-screen background and becomes its own panel in the middle of the column.

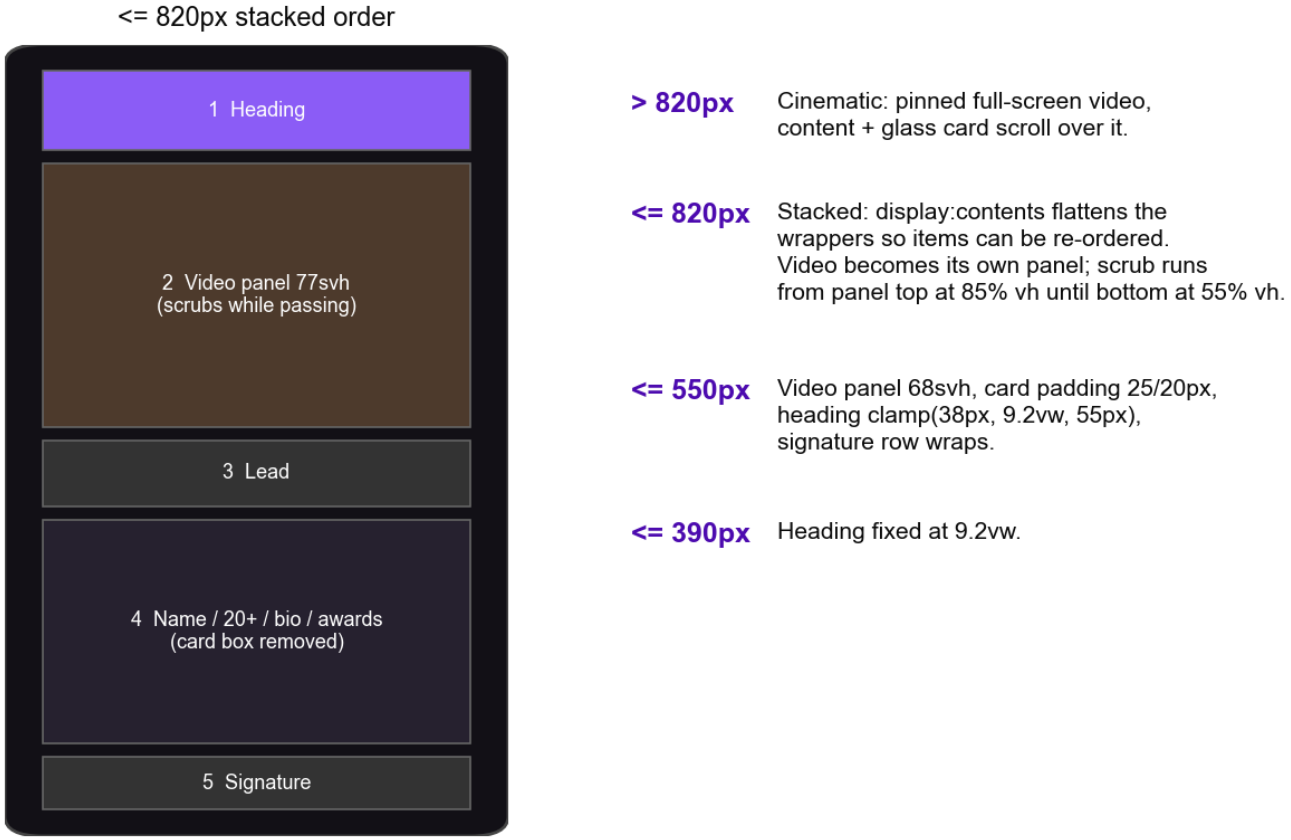


Figure 7.1 Small-screen order and what each breakpoint changes.

On small screens the scrub formula changes too: it measures the video panel instead of the whole section. The film starts when the panel's top is at 85% of the screen height and finishes while its last frames are still visible, so the whole film plays before the bio scrolls in.

Accessibility: with `prefers-reduced-motion` on, the reveal animations are switched off and the text is simply shown. The backdrop is `aria-hidden`; the headline is a real `h2` labelling the section.

8. The video asset

The video is the effect. The source uses a 7-second, 1280 x 720 clip of the founder with photo cards orbiting him. That clip is footage of a real person and their portfolio, so it is **not** copied into your frame: you supply your own clip (prompt file 02 helps you make one).

Requirements for a scrub-ready clip

Property	Recommended	Why
Length	5-8 s	the whole clip is spread over ~2.5 screens of scrolling; longer clips feel slow and weigh more
Resolution	1280x720 (desktop), optional 720x1280 mobile cut	it is shaded and partly covered; 1080p rarely shows a difference
Keyframes	every frame (-g 1)	see Figure 8.1; this is the single biggest factor for smoothness
Audio	none (-an)	muted anyway; saves size
Container	MP4 (H.264), +faststart	plays everywhere; metadata loads first so scrubbing can start early
Motion	slow, continuous camera push or orbit	scrolling both ways must look natural; avoid cuts and flashes
Composition	subject left-centre, dark right side	the card and lead sit on the right; text must stay readable
Poster	JPG of the first frame	shown before the video loads



Figure 8.1 With a normal export, jumping to a frame can require decoding up to 60 frames first, which causes stutter. With a keyframe on every frame, each seek costs one frame.

Encoding command

```
ffmpeg -i input.mp4 -vf "scale=1280:-2" -c:v libx264 -preset slow -crf 23 \
       -g 1 -pix_fmt yuv420p -an -movflags +faststart scroll.mp4

ffmpeg -i scroll.mp4 -frames:v 1 -q:v 3 poster.jpg
```

Trade-off: a keyframe-every-frame file is usually 2-4x larger than a normal export. For a 7 s 720p clip, aim for under ~10 MB; raise `-crf` (e.g. 26-28) if it is bigger.

9. Reusable package plan

Files

```
scroll-hero-frame/
  scroll-hero.css     all styles, .shf- prefix, themed by CSS variables
  scroll-hero.js      scrub + reveal, auto-starts every [data-scroll-hero]
  index.html          demo page (header + frame + spacer sections)
  README.md           how to use / customise
  assets/
    scroll.mp4        your scrub-ready clip   (you provide)
    poster.jpg        first frame            (you provide)
```

Changes from the source, and why

Source behaviour	Reusable version
Hard-coded IDs (#team, #founderScrollVideo)	Classes + data attribute, so several frames can live on one page
Generic class names (.team, .team-grid)	.shf- prefix, so it cannot clash with your project's CSS
Colours and sizes written inline in rules	CSS variables on the section: change one value to re-theme
Two separate scripts	One file, one init function per frame, plus a destroy() for single-page apps
Real name, awards, social links, video	Placeholder text and empty slots
Site header is global	Simple matching header included only in the demo page

Customisation variables

Variable	Default	Controls
--shf-accent	#4C0AAF	highlight colour everywhere
--shf-bg	#060508	section background
--shf-text	#F5F3F6	headline + name colour
--shf-min-height	175vh	minimum scroll length (longer = slower film)
--shf-max-width	1600px	content column width
--shf-pad-x	6.5vw	side padding
--shf-font-display	"Arial Black", Impact, Arial, sans-serif	big type
--shf-font-body	Arial, Helvetica, sans-serif	all other text
--shf-video-position	center 42%	which part of the video stays in view when cropped

Dropping it into another project

- Copy `scroll-hero.css` and `scroll-hero.js` into the project and link them.
- Paste the `<section class="shf" data-scroll-hero>` block where you want the frame; edit the text.
- Point the `<video>` at your encoded clip and poster.
- Optional: override variables, e.g. `.shf { --shf-accent: #e11d48; }`.
- React / Next / Vue: render the same markup and call `ScrollHero.init(el)` on mount, `destroy()` on unmount.

10. Risks, decisions and next steps

Risk / limit	Impact	Mitigation in the build
Clip not re-encoded	stuttering scrub	README + prompt 02 include the exact ffmpeg command
Large video file	slow first load on mobile	poster image, preload, keep clip short; optional separate mobile clip
iOS Safari won't draw paused video	black panel on iPhone	keep the source's play()/pause() priming on first touch
backdrop-filter cost	jank on low-end devices	card blur is only 10px on one element; safe fallback is the semi-opaque background
Very long text in the card	section grows; film plays slower	formula divides by real height, so timing stays correct
Copyrighted footage and identity	legal	frame ships with placeholders; you add your own media and text

Decisions you can make before the build

- Keep the demo text as the original wording, or use neutral placeholder copy?
- Include a separate portrait (9:16) video slot for phones, or one clip for all screens (as the source does)?
- Fonts: keep the system Arial Black (exact match), or allow a web font override?

Next steps

- You give the go-ahead.
- Build the four files; run the demo locally; compare screen by screen with your 8 screenshots.
- Deliver Report 02 (build) as PDF with side-by-side comparison and an updated build prompt in Markdown.

A. Glossary

Term	Meaning
Scrub / scrubbing	moving a video's playhead by hand (here: by scrolling) instead of letting it play
Sticky	an element that scrolls normally until it hits an edge of the screen, then stays there while its parent is still visible
Keyframe (GOP)	a full picture stored in the video; other frames only store changes. Seeking must start from the nearest keyframe
IntersectionObserver	browser feature that reports when an element enters or leaves the screen, without checking on every scroll
requestAnimationFrame	asks the browser to run code right before the next screen repaint (about 60 times a second)
clamp(a, b, c)	CSS: use value b, but never less than a or more than c
vh / svh / vw	1% of the viewport height / small-viewport height on phones (ignores the address bar) / width
backdrop-filter	blurs whatever is behind an element (the 'frosted glass' card)
display: contents	makes a wrapper vanish from layout while keeping its children, so they can be re-ordered
prefers-reduced-motion	a user setting asking sites to minimise animation; the frame honours it